# Assignment-4.3

TASK-1

Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

**Step 1: Find your router's IP address**

1. Open Command Prompt

Press Windows + R, type cmd, and press Enter.

2. Run the command

Type ipconfig and press Enter. Under your active Wi-Fi adapter, find Default

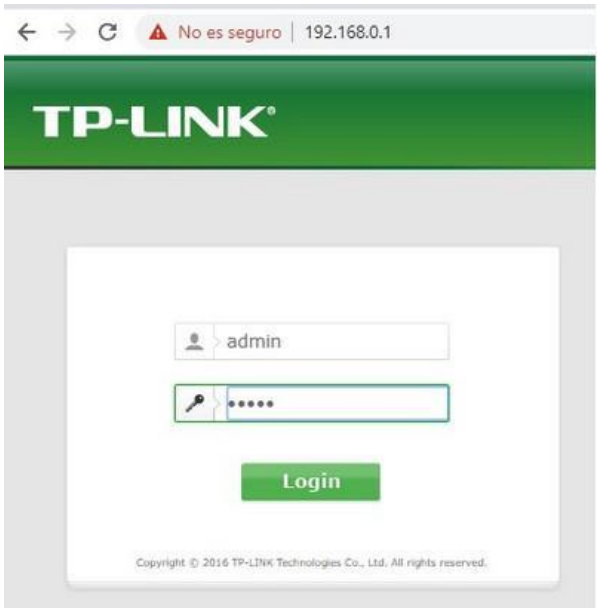
**Step 2: Open the router login page**

1.Connect your laptop to your home Wi-Fi.

2.Open Chrome or Edge.

3.Type your Default Gateway into the address bar, for example http://192.168.1.1.

4.Press Enter. If the router supports a secure login page, it may use HTTPS instead.




**Step 3: Take your screenshot**

When the router login page appears:

- Press Windows + Shift + S.
- Select the login page area.
- Save the screenshot using Snipping Tool or the notification that appears.

TASK-2

Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

1.Press Windows + R on your laptop.

2.Type cmd and press Enter.

3.Type ipconfig and press Enter.

4.Under Wireless LAN adapter Wi-Fi, find the required network details.

| **Parameter** | **Value** |
|---|---|
| IP Address (IPv4) | 10.141.249.240 |
| Subnet Mask | 255.255.255.0 |
| Default Gateway | 10.141.249.123 |

TASK-3

Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

**1. Gather Your Network Information**

Before changing settings, note down your current router details:

- **Router / Default Gateway IP:** e.g., 192.168.1.1
- **Subnet Mask:** e.g., 255.255.255.0
- **DNS Server:** e.g., 8.8.8.8 or your router IP

**2. Configure a Static IP on Your Device**

**Windows**

1. Open Settings > Network & Internet > Wi-Fi or Ethernet.
2. Click Hardware properties or Properties next to your network.
3. Look for IP assignment and click Edit.
4. Change the drop-down from *Automatic (DHCP)* to Manual.
5. Toggle IPv4 on.
6. Enter your chosen static IP (e.g., 192.168.1.50), Subnet mask (255.255.255.0), and Gateway (192.168.1.1).
7. Enter your Preferred DNS and click Save.

**3. Test Internet Access**

Open a web browser or run a ping test (e.g., ping 8.8.8.8 or ping google.com in your terminal/command prompt) to check if your internet connection works properly with the static IP.

**4. Revert to Automatic (DHCP)**

Go back to the same network settings menu on your device and switch the configuration back from *Manual / Static* to Automatic (DHCP) so your router can manage your IP addresses automatically and prevent future IP conflict issues.

TASK-4

In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Current DHCP Range: I could not check the range because I am using a mobile hotspot instead of a router admin panel. The range depends on the hotspot or router configuration.

What happens if two devices have the same IP address?

If two devices are assigned the same IP address, an IP address conflict occurs. Both devices may experience network connectivity problems, and they may be unable to access the internet properly.